Topic 2: User Behaviour (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

EMOTIONAL INTERACTION

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01 EMOTIONS AND BEHAVIOUR

02 EXPRESSIVE INTERFACES

03 AFFECTIVE COMPUTING & EMOTIONAL AI

04 PERSUASIVE TECHNOLOGY & BEHAVIOURAL CHANGE

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EMOTIONS & BEHAVIOUR

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Emotions and Behaviour

- HCI has traditionally been about designing efficient and effective systems
- Now more about how to design interactive systems that make people respond in certain ways
  - For example, to be happy, to be trusting, to learn, or to be motivated
- *Emotional interaction* is concerned with how we feel and react when interacting with technologies
- *Affective computing* is improving with better recognition software and machine learning algorithms

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Emotional Interaction

- What makes us happy, sad, annoyed, anxious, frustrated, motivated, delirious, and so on
  - Translating this into different aspects of the user experience
- Why people become emotionally attached to certain products (for instance, virtual pets)
- Can social robots help reduce loneliness and improve well-being?
- How to change human behaviour through the use of emotive feedback

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Activity



- Try to remember the emotions you went through when buying a big-ticket item online (for example, a refrigerator, a vacation, a computer)

- How many different emotions did you go through?

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Why has this simple way of obtaining visitor feedback been so effective?



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Pulling at the heart strings with an emotive message



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Pulling at the heart strings with an emotive message



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Is it possible to design an interface to match or change how we are feeling?

- Should an interface be designed to improve how we feel?
  - If so, how?
- Our moods and feelings are continuously changing
  - How does the interface keep track and know when to do something?
- What moods match which kinds of interfaces?
- How would you design an interface for when someone is happy, angry, sad, bored, or focused?

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How do emotions affect behaviour and vice versa?

- Does being angry make you concentrate better or more distracted?
- When you are happy do you take more risks such as spend more money or buy more?
- Baumeister et al (2007) argue the relationship is more complex than a single cause-and-effect model

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Automatic (affect) vs. Conscious Emotions

- Emotions can be short-lived (for instance, a fit of anger) or complex and long-lasting (for example, jealousy)
- Emotions have been categorised as automatic or conscious:
  - Automatic ones are rapid and dissipate quickly
  - Conscious ones develop slowly and take a long time to go (for instance, reflection)

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Designing with the 3-levels in mind

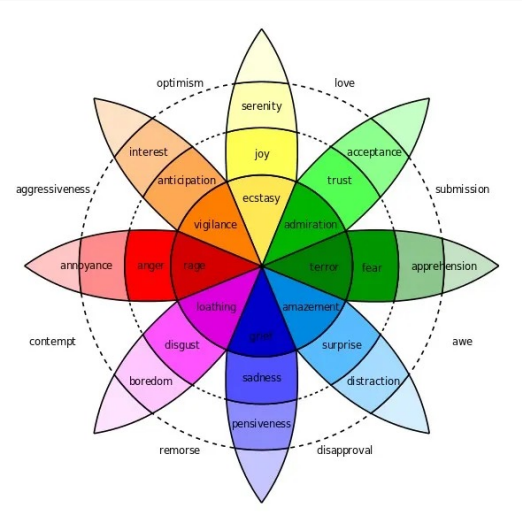
- Visceral design refers to making products look, feel, and sound good
- Behavioural design is about use, and it equates with traditional values of usability
- Reflective design is about considering the meaning and personal value of a product

Analysing a Swatch watch design using the model



- Cultural images and graphical elements designed at the reflective level
- Affordances of use at the behavioral level
- Brilliant colours and wild design attract user's attention at the visceral level

Plutchik's (1980) Wheel of Emotions



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Plutchik's Wheel

- Categorises human emotions into 7 well known emotions
  - Anger, Disgust, Fear, Sadness, Anticipation, Joy and Surprise and Trust (not usually considered an emotion)
- Emotional labels are added to these
  - optimism, love, submission, awe, disproval, remorse, contempt, aggressiveness
- Colours used in the wheel reflect the intensity of an emotion
  - the darker the shade, the more intense the emotion is
- Can be used as a 'colour palette' akin to a UX mood board
  - elicit different kinds and levels of emotional response
- It does not provide prescriptive advice on how to instruct design for a selection of emotions

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EXPRESSIVE INTERFACES

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Expressive Interfaces

- Provide reassuring feedback that can be both informative and fun
- Can also be intrusive, however, causing people to become annoyed and even angry
- Colour, icons, sounds, graphical elements, and animations are used to make the 'look and feel' of an interface appealing
  - Conveys an emotional state
- In turn, this can affect the usability of an interface
  - People are prepared to put up with certain aspects of an interface (for instance, slow download rate) if the end result is appealing and aesthetic

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The Appearance of an Interface

(a) Emotional icons were used in the 1980s to indicate rebooting or crashed computer

- Smiling apple face

(b) Nowadays, computers use more impersonal but aesthetically-pleasing icons to indicate that the user needs to wait

- Beachball



(a)



(b)

The Design of Thermostats

(a) The Nest thermostat has a minimalist and aesthetically-pleasing design

- Round face and simple dial
- Large font and numbers

(b) It is very different from earlier thermostat designs

- Utilitarian and dull



(a)



(b)

Design of Website to be Aesthetics



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AFFECTIVE COMPUTING & EMOTIONAL AI

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Affective Computing & Emotional AI

- Affective computing is concerned with how to use computers to recognise and express emotions as humans do (Picard, 1998)
- It involves designing ways for people to communicate their emotional state
- It uses sensing technologies to measure GSR, facial expressions, gestures, and body movement
- Explores how affect influences personal health
- *Emotional AI* aims to automate the measurement of feelings and behavior using AI to infer them from facial expressions and voice
- The goal is to predict user's emotions and aspects of their behavior
  - For example, what is someone most likely to buy online when feeling sad, bored, or happy

Techniques Used

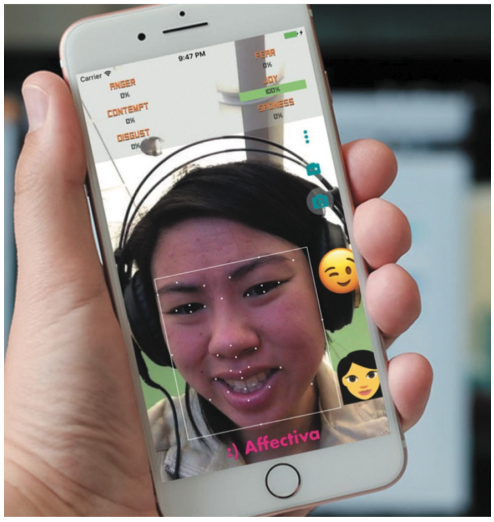
- Cameras for measuring facial expressions
- Biosensors placed on fingers or palms to measure GSR
- Affective expression in speech (for example, intonation, pitch, and loudness)
- Body movement and gestures using accelerometers and motion capture systems

Classification of Emotions

- Six core expressions typically measured:
  - Sadness, disgust, fear, anger, contempt, and joy

- Type of facial expression chosen by AI through detecting presence of absence of:
  - Smiling, eye widening, brow raising, brow furrowing, raising a cheek, mouth opening, upper-lip raising, and wrinkling of the nose

Facial Coding using Affdex software

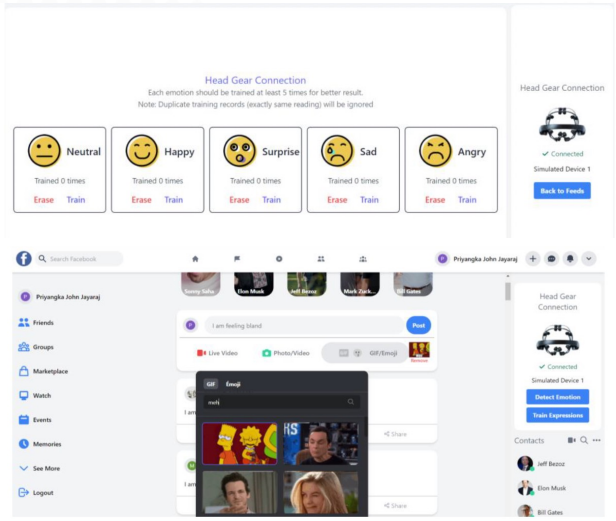


How is this Emotional Data used?

- If user screws up their face when an ad pops up
  - ➡ Feel disgust
- If user starts smiling
  - ➡ They are feeling happy
- Website can adapt its ad, movie storyline, or content to match user's emotional state
- If system used in a car, it might detect an angry driver and suggest they take a deep breath
- Eye-tracking, finger pulse, speech, and words/phrases also analyzed when tweeting or posting to Facebook

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Detecting & Expressing Emotions





Publication:

Priyangka John Jayaraj and Masitah Ghazali. User's Perspectives on Expressing Emotions on Social Media Platforms with Mobile Brain-Computer Interface. In Fusion 2021 Proceedings of 3rd National Symposium on Human-Computer Interaction, 19th August 2021, pp. 8-17. eISBN: 978-967-18511-1-1

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Indirect Emotion Detection

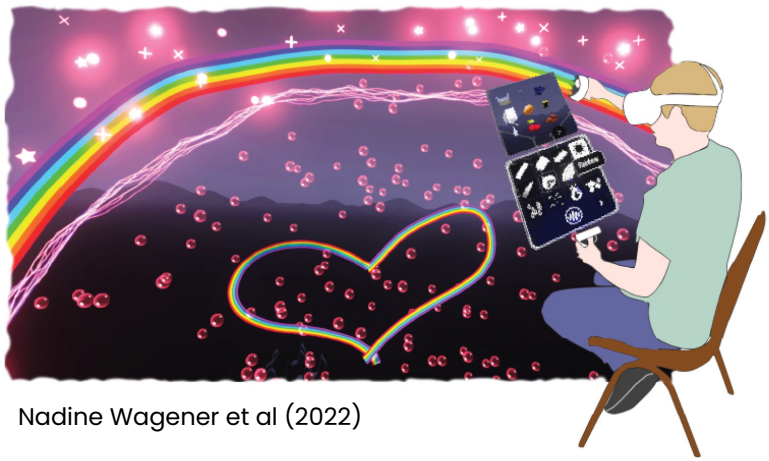
- Also used to infer or predict someone's behaviour
  - e.g, determining a person's suitability for a job or how they will vote in an election

- Do you think it is ethical that technology can read your emotions from your facial expressions or from your tweets?

Detecting and Reflecting on Moods

- How can technology be designed to help people understand more about their moods and mood swings?
- We can be in a good mood one day and then a bad mood the next day. How does this happen and why?
- Not possible to detect different moods in the way tech is used for emotion detection
  - often not expressed through obvious physiological responses
- Mobile mood tracker apps (e.g., Moodnotes, Daylio)
  - intended to help people keep track of their moods
  - to reflect more on why they might be feeling gloomy or cheerful
  - understanding their moods is assumed to help improve well-being
- Virtual reality is also used to enable people to explore their moods (e.g. Mood Worlds)

Mood Worlds: a VR exploratory tool



Nadine Wagener et al (2022)

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PERSUASIVE TECHNOLOGIES & BEHAVIOURAL CHANGE

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Persuasive Technologies & Behavioural Change

- Interactive computing systems designed to change people's attitudes and behaviors (Fogg, 2003)
- A diversity of techniques now used to change what they do or think
  - Pop-up ads, warning messages, reminders, prompts, personalized messages, recommendations, or Amazon 1-click
  - Commonly referred to as *nudging*

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Nintendo's Pocket Pikachu

- Developed to change bad habits and improve well being
  - Designed to motivate children to be more physically active on a regular basis
  - Owner of the digital pet that 'lives' in the device is required to walk, run, or jump
  - If owner does not exercise, the virtual pet becomes angry and refuses to play anymore

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How Effective?

- Can interactive technologies that monitor, nag, or behave like a human keep them interested in looking after it and in doing so make themselves more fit?
- How does looking after a virtual pet change a child's behavior?
  - Emotional attachment
  - Happy Pokemon makes them feel good
  - Sulking Pokemon makes them feel bad



Dilemma: Should voice assistants teach kids good manners?

- Many children talk to Alexa as if she was their friend
- They also learn that it is not necessary to say **please** and **thank you** to her when asking questions
- Is this lack of using etiquette a problem?
- Would it transfer over to real life situations?
  - For example, demanding "Auntie, get me my drink."
- Should parents or voice assistants teach their kids good manners
- How much parental control should voice assistants be given?
  - Would children find it weird or creepy that their Alexa (who is their friend) nags them to clean their teeth?
- Recent research has shown that children know to treat humans differently compared with the way they talk to a voice assistant (Alexis Hiniker et al., 2021)

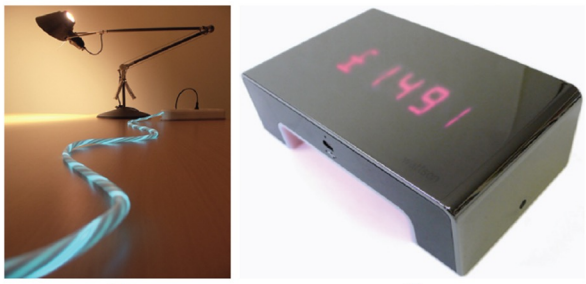
Tracking Devices

- Mobile apps designed to help people monitor and change their behaviour (for instance, fitness, sleeping, or weight)
- Can compare with online leader boards and charts to show how they have done in relation to their peers and friends
- Also apps that encourage reflection, which in turn increase well-being and happiness

Sustainable HCI

- Focus on designing tech interventions to help people reduce their energy consumption
- An effective technique is to provide homeowners with feedback on their consumption
- Simple infographics and emoticons are often most powerful:
  - Encourage people to reflect and see what they can change to reduce their energy consumption
- Peer pressure and social norms are also powerful methods

Energy Reduction



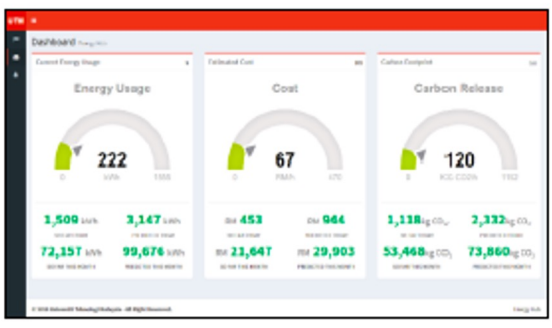
(a) (b)

Figure 5.12 (a) The Power Aware Cord consists of an electrical power strip in which the cord is designed to visualize the energy rather than hiding it. Increase and decrease in use is conveyed through showing glowing pulses, flow, and intensity of light. (b) The Waatson (now a commercial product available in many countries) measures in watts or cost how much electricity someone is using in their home at any moment. This is conveyed in LEDs on the top side. On the underside are colored lights: when they glow blue it shows you are using less than normal; when it changes to purple it indicates that your usage is average: and when it is red it indicates you are using more than normal

Source: (a) Photo taken from the Interactive Institute's research program "Static!" and reproduced with permission. (b) Reproduced with permission from DIY Kyoto Ltd. www.diykyoto.com.

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Energy Reduction via Peripheral Display





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The Tidy Street Project

- Large-scale visualization of the street's electricity usage
  - Stenciled display on the road surface using chalk
  - Provided real-time feedback that all could see change each day
  - Reduced electricity consumption by 15 percent

(Bird and Rogers, 2010)



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Phishing Scams

- Web used to deceive people into parting with personal details
  - For example, PayPal, eBay, and "you won the lottery" emails
- Allows Internet fraudsters to access their bank accounts and draw money from them
- Many vulnerable people fall for it
- The art of deception is centuries old but internet allows ever more ingenious ways to trick people

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Summary

- Emotional aspects of interaction design are concerned with how to facilitate certain states (for example, pleasure) or avoid reactions (for instance, frustration)
- Well-designed interfaces can elicit good feelings in people
- Aesthetically-pleasing interfaces can be a pleasure to use
- Expressive interfaces can provide reassuring feedback to users
- Badly designed interfaces make people frustrated, annoyed, or angry
- Emotional AI and affective computing use AI and sensor technology for detecting people's emotions by analyzing their facial expressions and conversations
- Emotional technologies can be designed to persuade people to change their behaviors